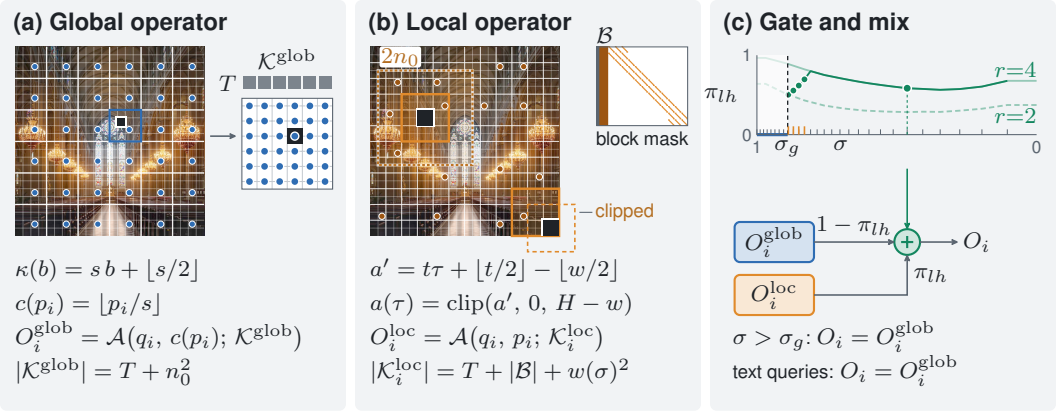
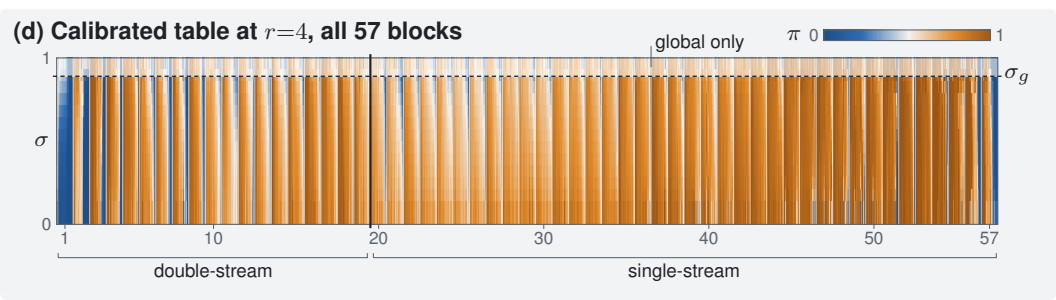

(a) Global operator
$\mathcal{K}^{\text{glob}}$
T
$\kappa(b) = s\,b + \lfloor s/2 \rfloor$
$c(p_i) = \lfloor p_i / s \rfloor$
$O_i^{\text{glob}} = \mathcal{A}(q_i,\, c(p_i);\, \mathcal{K}^{\text{glob}})$
$|\mathcal{K}^{\text{glob}}| = T + n_0^2$
(b) Local operator
$2n_0$
$\mathcal{B}$
block mask
clipped
$a' = t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor$
$a(\tau) = \text{clip}(a',\, 0,\, H - w)$
$O_i^{\text{loc}} = \mathcal{A}(q_i,\, p_i;\, \mathcal{K}_i^{\text{loc}})$
$|\mathcal{K}_i^{\text{loc}}| = T + |\mathcal{B}| + w(\sigma)^2$
(c) Gate and mix
π_{lh}
$r{=}4$
$r{=}2$
σ_g
σ
O_i^{glob}
$1 - \pi_{lh}$
O_i
O_i^{loc}
$\sigma > \sigma_g$: $O_i = O_i^{\text{glob}}$
text queries: $O_i = O_i^{\text{glob}}$


(d) Calibrated table at $r{=}4$, all 57 blocks
global only
π
σ
σ_g
double-stream
single-stream